Type MS · Where two lines meet

Answer 1 of 4 · Question 2 part b · 3 marks

2023 JAN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

2 a	(Gradient of AB =) $\frac{0-3}{4+5}\left(=-\frac{1}{3}\right)$	M1
	Gradient of the perpendicular = $-\frac{1}{"-\frac{1}{3}"}(=3)$	M1
	$y-5=-\frac{1}{"-\frac{1}{3}"}(x+1)$ or $y-5="3"(x+1)$ or $y=3x+8$	M1
	$3x-y+8=0$ or $-3x+y-8=0$ oe	A1 (4)
b	(Equation of AB =) $y-0="-\frac{1}{3}"(x-4)$ or $y=-\frac{1}{3}x+\frac{4}{3}$ oe	M1
	Solve $y-3x-8=0$ and $y-0=-\frac{1}{3}(x-4)$ oe simultaneously	M1
	$0=\frac{10}{3}x+\frac{20}{3}$ oe	
	$x=-2$ $y=2$ or (-2, 2) *	A1 cso (3)
	Alternative for final 2 marks (substitutes (- 2, 2) into each equation))	
	Line l is $3x-y+8=0$	
	When $x=-2$ and $y=2$, $-6-2+8=0\Rightarrow D$ lies on l	
	AB is $y=-\frac{1}{3}\ x-4$	{M1}
	When $x=-2$ and $y=2$, $-\frac{1}{3}\ -2-4\ =2=y\Rightarrow D$ lies on AB	
	So l and AB intersect at $-2, 2$ *	{A1 cso} (3)
c	Midpoint of AB = $\left(\frac{-5+4}{2},\frac{3(+0)}{2}\right)=\left(-\frac{1}{2},\frac{3}{2}\right)$	B1 (M1 on ePen)
	$\neq(-2,2)$ So not perpendicular bisector of AB *	B1 (A1 on ePen) cso (2)
ALT	$BD=\sqrt{(4--2)^2+(0-2)^2}\left(=\sqrt{40}\right)$ or $AD=\sqrt{(-5--2)^2+(3-2)^2}\left(=\sqrt{10}\right)$	M1
	$BD=\sqrt{40}(=2\sqrt{10})$ and $AD=\sqrt{10}$	A1cso
	$BD\neq AD$ so not the perpendicular bisector	(2)

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d	$BD = \sqrt{(4--2)^2+(0-2)^2}\ (=\sqrt{40})$ oe	M1
	$CD = \sqrt{(-1--2)^2+(5-2)^2}\ (=\sqrt{10})$ oe	M1
	$(\tan\angle ABC =)\ \dfrac{"\sqrt{(-1--2)^2+(5-2)^2}"}{"\sqrt{(4--6)^2+(0-2)^2}"}$ or $\dfrac{\sqrt{10}}{\sqrt{40}}$	ddM1
	$=\dfrac{1}{2}$	A1 (4)
ALT 1	Uses cosine rule in triangle ABC	
	$AB = \sqrt{(-5-4)^2+(3-0)^2}\ (=\sqrt{90}=3\sqrt{10})$ or $BC = \sqrt{(4--1)^2+(0-5)^2}\ (=\sqrt{50}=5\sqrt{2})$ or $AC = \sqrt{(-1--5)^2+(5-3)^2}\ (=\sqrt{20}=2\sqrt{5})$	M1
	may also find expressions for $(AB)^2, (BC)^2$ or $(AC)^2$	
	$AB = \sqrt{(-5-4)^2+(3-0)^2}\ (=\sqrt{90}=3\sqrt{10})$ and $BC = \sqrt{(4--1)^2+(0-5)^2}\ (=\sqrt{50}=5\sqrt{2})$ and $AC = \sqrt{(-1--5)^2+(5-3)^2}\ (=\sqrt{20}=2\sqrt{5})$	M1
	may also find expressions for $(AB)^2, (BC)^2$ and $(AC)^2$	
	$(\cos ABC =)\ \dfrac{("\sqrt{50}")^2+("\sqrt{90}")^2-("\sqrt{20}")^2}{2\times"\sqrt{50}"\times"\sqrt{90}"}\left(=\dfrac{2}{\sqrt{5}}=\dfrac{2\sqrt{5}}{5}\right)$	ddM1
	$(\tan ABC)=\dfrac{1}{2}$	A1
		Total 13 marks

Type MS · Where two lines meet

Answer 1 of 4 · Question 2 part b · 3 marks

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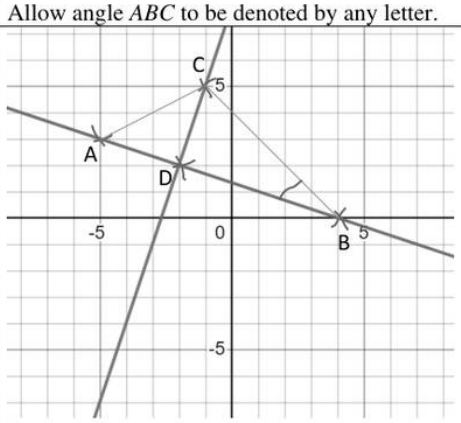
Part	Mark	Additional Guidance
(a)	M1	For correctly finding the gradient of AB, need not be simplified.
	M1	For finding the negative reciprocal of their gradient of AB, need not be simplified.
	M1	For a fully correct method using their negative reciprocal to find the equation of the straight line. Allow use of their $-\frac{1}{"-\frac{1}{3}"}$ which can be unsimplified or processed incorrectly, but must follow from a clear attempt to have found the negative reciprocal of their gradient for AB. If $y = mx + c$ is used, there must be a fully correct substitution and a fully correct rearrangement to find c
	A1	For $3x - y + 8 = 0$ or $-3x + y - 8 = 0$ oe so long as integer coefficients and = 0.
(b)	M1	For writing the equation of AB, the equation need not be simplified, but must be fully correct.
	M1	For an attempt to solve their equations simultaneously, allow one processing error. Must eliminate one variable and arrive at a value for x or y.
	A1 cso*	For obtaining the required result, no errors.
	ALT	**Final 2 marks**
	M1	For writing the equation of AB, the equation need not be simplified, but must be fully correct.
	M1	For a fully correct substitution of (–2, 2) into a correct equation for AB to show that D lies on AB **OR** a fully correct substitution of (–2, 2) into **their** equation for line l
	A1 cso*	For a fully correct substitution of (–2, 2) into a correct equation for AB to show that D lies on l **AND** a fully correct substitution of (–2, 2) into a correct equation for line l to show that D lies on l. An appropriate (minimal) conclusion must also be written, this can be as simple as # or 'shown'. No errors.
(c)	B1 (M1 on ePen)	For either coordinate of the midpoint correct
	B1 (A1 on ePen) cso*	For both coordinates correct and an appropriate (minimal) conclusion, which must be correct, can be as simple as # or 'shown'.
ALT	M1	For a correct method to calculate AD or BD
	A1 cso*	For correct values for AD and BD, accept decimals and a (minimal) conclusion, can be as simple as # or 'shown'.
(d)	M1	For the correct method to calculate BD (May be seen in (c))
	M1	For the correct method to calculate CD
	ddM1	For $\left(\tan\angle ABC =\right)\frac{\sqrt{(-1--2)^2+(5-2)^2}}{\sqrt{(4--6)^2+(0-2)^2}}$ or $\frac{\sqrt{10}}{\sqrt{40}}$ allow their BD and CD. Dependent on the previous 2 method marks.
	A1	For $\left(\tan\angle ABC\right)=\frac{1}{2}$
If you see a method using the gradient of BC *and* tan $(A - B)$ – PLEASE SEND TO REVIEW		

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Type MS · Where two lines meet

Answer 1 of 4 · Question 2 part b · 3 marks

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ALT 1	**M1**	Correct method to calculate one of the required lengths
	M1	Correct method to calculate all of the required lengths
	ddM1	Correct substitution into the cosine rule to give the expression shown, allow use of their lengths. Must obtain a correct expression for cosABC = , using their values. Dependent on the previous 2 method marks.
	A1	For $(\tan \angle ABC) = \frac{1}{2}$
Because of the wording of the question, in not specifically asking for an exact value, we are going to allow a range for the value of $\tan \angle ABC$ as follows: $0.49989 \leq \tan \angle ABC \leq \frac{1}{2}$		
If you see the candidate using the ratio for a right angled triangle - but they are using the wrong pairing of lengths, (perhaps AB, BC or AC) then this is to be marked using the main scheme, and is unlikely to be awarded any marks unless they have correctly calculated BD or CD. To be clear - If the candidate states correct values for AB, BC or AC but goes onto to use a tan ratio - this cannot be awarded any marks using the ALT mark scheme, because they are not using the correct cosine rule and with the tan ratio they should be using at least one of the correct sides, BD or CD. If you see a clear attempt to use the cosine rule, please use this ALT marking scheme. A special case is when the candidate has not made it clear which route they were intending to take - tan ratio (right angled triangle ratio) or cosine rule. If they have all lengths stated correctly, they could be awarded M1 M1 using either scheme.		
Further ALTS		If you see a valid method, leading to a fully correct exact value for tanABC, please award 4 marks. **If the value for tanABC is incorrect or approximate and there is some partially correct working, PLEASE SEND TO REVIEW.** Allow angle ABC to be denoted by any letter.
		 For reference $BD = \sqrt{40} (= 2\sqrt{10})$ $AD = \sqrt{10}$ $AB = \sqrt{90} = 3\sqrt{10}$ $BC = \sqrt{50} = 5\sqrt{2}$ $\sin ABC = \frac{\sqrt{5}}{5}$ $\cos ABC = \frac{2\sqrt{5}}{5}$

Type MS · Where two lines meet

Answer 1 of 4 · Question 2 part b · 3 marks

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Question number	Scheme	Marks
	a	M1 A1

Type MS · Where two lines meet

Answer 2 of 4 · Question 11 part d · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 4 · find part d in this mark scheme extract

Number		
11 (a)	$\frac{y-6}{x-5}=\frac{6-3}{5--1}\Rightarrow 2y-12=x-5$	M1A1 [2]
(b)	$\left(\frac{2\times5+1\times-1}{2+1},\frac{2\times6+1\times3}{3}\right)\Rightarrow(3,\ 5)*$	M1A1 [2]
(a)		
M1	A fully correct method for finding the equation of a straight line e.g. $\frac{y-y_1}{x-x_1}=\frac{y_2-y_1}{x_2-x_1}$	
A1	$2y-12=x-5$ oe	
(b)		
M1	Use of $\left(\frac{qx_1+px_2}{p+q},\frac{qy_1+py_2}{p+q}\right)$ or $\begin{pmatrix}-1\\3\end{pmatrix}+\frac{2}{3}\times\begin{pmatrix}6\\3\end{pmatrix}$	
A1	Obtains the **given** coordinates	

Type MS · Where two lines meet

Answer 2 of 4 · Question 11 part d · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 4 · find part d in this mark scheme extract

(c)	Gradient of perpendicular to AB = -2	B1
	$\frac{5-n}{3-m}=-2 \Rightarrow (11=2m+n)$	M1
	Radius = 5, so the length of AC = 10	
	$100=(3-n)^2+(-1-m)^2 \Rightarrow (90=n^2-6n+m^2+2m)$	M1
	Solves simultaneous equations $90=121-44m+4m^2-6(11-2m)+m^2+2m$	depM1
	$35=5m^2-30m \Rightarrow m^2-6m-7=0$	depM1
	$(m-7)(m+1)=0 \Rightarrow m=7,\ n=-3$	A1 [6]
ALT (c) using vectors		
(c)	$\overrightarrow{AB}=\begin{pmatrix}5\\6\end{pmatrix}-\begin{pmatrix}-1\\3\end{pmatrix}=\begin{pmatrix}6\\3\end{pmatrix} \Rightarrow$ Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$	B1
	$\lvert\overrightarrow{AB}\rvert=\sqrt{45}=3\sqrt{5} \Rightarrow \lvert AP\rvert=2\sqrt{5}$ $\quad \lvert\overrightarrow{AC}\rvert=10$	M1M1
	Using Pythagoras $\lvert\overrightarrow{PC}\rvert=\sqrt{100-20}=\sqrt{80}=4\sqrt{5}$	depM1
	$\overrightarrow{PC}=\frac{4\sqrt{5}}{3\sqrt{5}}\times\begin{pmatrix}3\\-6\end{pmatrix}=\begin{pmatrix}4\\-8\end{pmatrix}$	depM1
	$\Rightarrow$ Coordinates of C are $(3+(4),\ 5+(-8)) \Rightarrow (7,-3)$	A1 [6]

(c)

B1 Gradient of perpendicular to AB = -2 (Can be implied by M1)

M1 Obtains an equation using the gradient of the perpendicular and the points (m, n) and P (Condone if given in terms of x and y)

M1 Obtains a second equation using AC = 10 and the points A and C

depM1 Solve simultaneously to obtain a 3 term quadratic

depM1 Solve their 3 term quadratic

A1 $m = 7,\ n = -3$

ALT:

B1 Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$

M1 Finds $\lvert\overrightarrow{AB}\rvert$ or $\lvert\overrightarrow{AP}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\lambda\begin{pmatrix}3\\-6\end{pmatrix}$

M1 States $\lvert\overrightarrow{AC}\rvert=10$ or $\lambda=\frac{4}{3}$

depM1 Use of Pythagoras to find $\lvert\overrightarrow{PC}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\frac{4}{3}\times\begin{pmatrix}3\\-6\end{pmatrix}$

depM1 Finds $\overrightarrow{PC}$

A1 $m = 7,\ n = -3$

Type MS · Where two lines meet

Answer 2 of 4 · Question 11 part d · 3 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 4 · find part d in this mark scheme extract

(d)	$\frac{y-'-3'}{x-'7'}=\frac{1}{2}\Rightarrow\left\{y=\frac{x-13}{2}\right\},\ \frac{y-3}{x--1}=-2\Rightarrow\{y=-2x+1\}$ Solving simultaneous equations by any method $\frac{x-13}{2}=-2x+1\Rightarrow p=3$ and $q=-5$	M1 M1A1 [3]
ALT (d) using vectors		
(d)	$\overrightarrow{AD}=\overrightarrow{PC}=\begin{pmatrix}4\\-8\end{pmatrix}$ Coordinates of point D $\Rightarrow(-1+(4),\ 3+(-8))=(3,-5)$	M1 M1A1 [3]
(e)	Length of AB $\sqrt{(6-3)^2+(5--1)^2}=3\sqrt{5}$ Length of CD $\sqrt{(-3--5)^2+(7-3)^2}=\sqrt{20}=2\sqrt{5}$ Area of trapezium $=\frac{1}{2}(3\sqrt{5}+2\sqrt{5})\times 4\sqrt{5}=50$ (units^2)	M1 A1 M1A1 [4]
ALT (e) using vectors		
	Area $=\frac{1}{2}\begin{vmatrix}-1 & 5 & 7 & 3 & -1\\ 3 & 6 & -3 & -5 & 3\end{vmatrix}$ $=\frac{1}{2}[(-6-15-35+9)-(15+42-9+5)]=50$	 M1A1 M1A1 [4]
		Total 17 marks

Type MS · Where two lines meet

Answer 2 of 4 · Question 11 part d · 3 marks

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Type MS · Where two lines meet

Answer 3 of 4 · Question 6 part b · 5 marks

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Number		
6 (a)	$\frac{\text{Change in } y}{\text{Change in } x} = \frac{2-0}{2-3} = -2$	M1 A1
	$2 = 1 + c$	M1
	$c = 1$	A1 ft
	$x - 2y + 2 = 0$	A1
		(5)
(b)	$2y - 2 = 7y + 3$	M1
	$-5y = 5$	
	$y = -1$	A1
	When $y = -1$	
	$x = 2 \times -1 - 2 = -4$ So $C = (-4, -1)$	A1
	$\left(\frac{3-4}{2}, \frac{0-1}{2}\right) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$	M1A1
		(5)
(c)	$AB = \sqrt{5}$	M1
	$BC = \sqrt{45}$	A1
	Area $= \frac{1}{2} \times \sqrt{5} \times \sqrt{45} \times \frac{1}{2}$	M1
	3.75	A1
		(4)
	Alternative c	**(14)**
	$\pm\frac{1}{2}\begin{vmatrix} 3 & 0 & 1 \\ 2 & 2 & 1 \\ -4 & -1 & 1 \end{vmatrix}$	M1
	$\pm\frac{1}{2}\left[3\begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ -4 & -1 \end{vmatrix}\right]$	A1
	$\pm\frac{1}{2}[3(2+1) + (-2+8)] \times \frac{1}{2}$	M1
	3.75	A1

Part	Mark	Additional Guidance
(a)	M1	For an attempt to find the gradient using the given coordinates **and** a correct attempt to find the perpendicular gradient. Accept either $\frac{2-0}{2-3} = (-2)$ or $\frac{0-2}{3-2} = (-2) \Rightarrow m_p = -\frac{1}{'-2'}$
	A1	For $m = \frac{1}{2}$
	dM1	For a correct method to find the equation of a line $y - 2 = '\frac{1}{2}'(x-2)$ or $y - 0 = '\frac{1}{2}'(x-3)$ **The gradient must come from a correct attempt to find the gradient and the gradient of the perpendicular** If $y = mx + c$ is used, then they must use the correct values of x and y and a value for c must be reached before this mark is awarded.
	A1	For the correct equation in any form $y - 2 = \frac{1}{2}(x-2)$ or $y - 0 = \frac{1}{2}(x-3)$ or $y = \frac{1}{2}x + 1$ oe

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Answer 3 of 4 · Question 6 part b · 5 marks

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	A1	For the correct equation in the required form $x-2y+2=0$ **oe** arranged in any order but all one side (e.g. accept even $\frac{x}{2}-y+1=0$)	
(b)	M1	Sets $L_1=L_2$ **and** attempts to solve for y or x	
		$2y-2=7y+3$ $-5y=5\Rightarrow y=...$	$\frac{x+2}{2}=\frac{x-3}{7}\Rightarrow 7x+14=2x-6$ $5x=-20\Rightarrow x=...$
	A1	$y=-1$	$x=-4$
	A1	$x=-4$	$y=-1$
	M1	For any **correct** method to find the coords of M using **their** values for C of x and y and the given coordinates of A (3, 0) $\left(\frac{3+['-4']}{2},\frac{0+['-1']}{2}\right)$ **This is a B mark in Epen**	
	A1	$\left(-\frac{1}{2},-\frac{1}{2}\right)$ **This is a B mark in Epen**	
(c)	M1	For attempting to find the length AB **and** BC $AB=\sqrt{(3-2)^2+(0-2)^2}$ and $BC=\sqrt{(2--4)^2+(2--1)^2}$ **This is a B mark in Epen**	
	A1	For both $AB=\sqrt{5}$ and $BC=\sqrt{45}$ **This is a B mark in Epen**	
	M1	$\frac{1}{2}\left('\sqrt{5}'\times'\sqrt{45}'\right)\times\frac{1}{2}$	For using a correct method to find the area of the triangle using correct lengths. i.e. they must be using BC and AB
	A1	For $A=3.75$	
	ALT – using determinants		
	M1	For using a correct method with their coordinates for C in any order (it is a triangle), but they must start and finish with the same coordinates $A=\frac{1}{2}\begin{pmatrix}3 & 2 & '-\frac{1}{2}' & 3\\ 0 & 2 & '-\frac{1}{2}' & 0\end{pmatrix}$ **This is a B mark in Epen**	
	A1	For using the correct coordinates $A=\frac{1}{2}\begin{pmatrix}3 & 2 & -\frac{1}{2} & 3\\ 0 & 2 & -\frac{1}{2} & 0\end{pmatrix}$ **This is a B mark in Epen**	
	M1	For a correct evaluation using their coordinates $A=\frac{1}{2}\left(\left[3\times2+2\times'-\frac{1}{2}'+'-\frac{1}{2}'\times0\right]-\left[2\times0+'-\frac{1}{2}'\times2+3\times'-\frac{1}{2}'\right]\right)=...$	

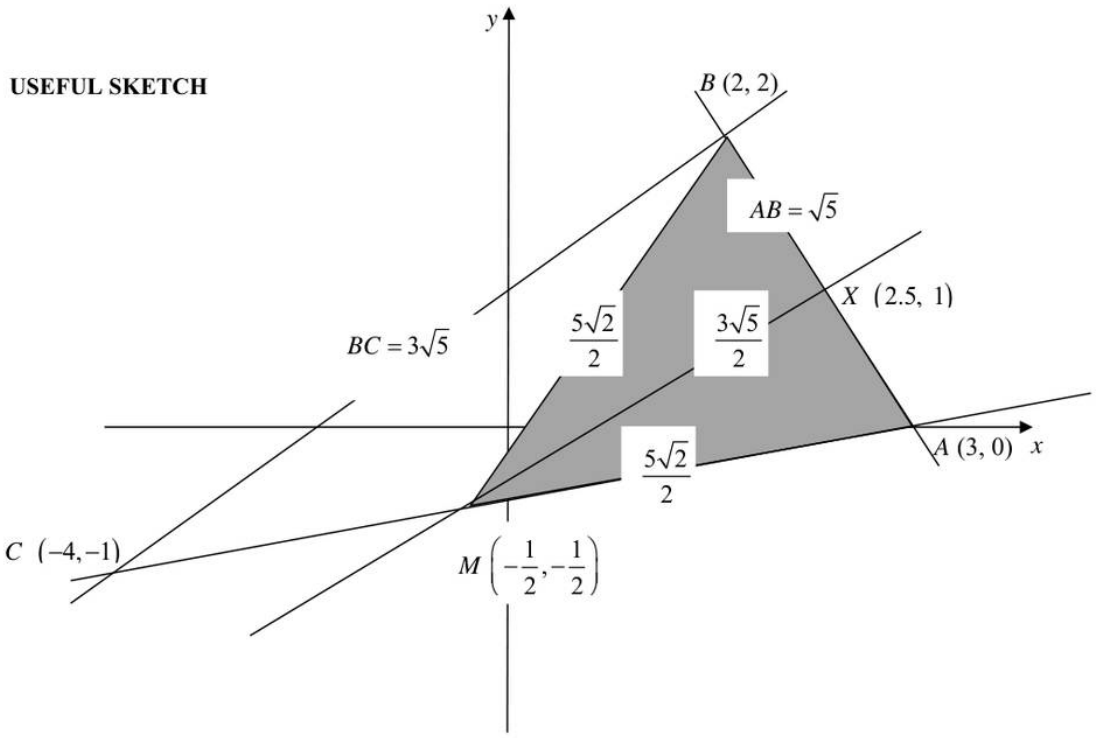
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	A1	For $A = 3.75$
	ALT	
	M1	For finding the length $AB = \sqrt{(3-2)^2 + (0-2)^2}$ **and** $MX = \frac{1}{2}\sqrt{3^2 + \left(\frac{3}{2}\right)^2}$ (Let X be midpoint of AB so MX is height of triangle ABM)
	A1	$AB = \sqrt{5}$ $MX = \frac{3\sqrt{5}}{2}$
	M1	Area of $\Delta ABM = \frac{1}{2} \times AB \times MX = \frac{1}{2} \times '\sqrt{5}' \times '\frac{3\sqrt{5}}{2}' = \left(\frac{15}{4}\right)$
	A1	Area of $\Delta ABM = \frac{15}{4} = 3.75$
If they use trigonometry, please send to review		

USEFUL SKETCH



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Answer 3 of 4 · Question 6 part b · 5 marks

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Question number	Scheme	Marks

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Answer 4 of 4 · Question 9 part a · 7 marks

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9 a	$L_1 : y-7=m(x-4)$	B1
	Gradient of $L_2 = -\frac{1}{m}$	M1
	$L_2 : y-k=-\frac{1}{m}(x-4)$	M1
	$x=\frac{y-7}{m}+4$	M1
	$y-k=-\frac{1}{m}\left(\frac{y-7}{m}+4-4\right)$	M1
	$m^2y-m^2k=7-y$	
	$y\left(m^2+1\right)=7+m^2k$	dM1
	$Y=\frac{7+m^2k}{m^2+1}$ *	A1 cso (7)
b	Midpoint of $AB=\left([4],\frac{k+7}{2}\right)$	B1
	So $\frac{k+7}{2}=\frac{7+m^2k}{m^2+1}$	M1
	$m^2k+7m^2+k+7=2m^2k+14$	
	$m^2\left(k-7\right)=k-7$	M1
	As $k\neq 7$ $m=\pm 1$	A1
	As $m<0$ $m=-1$	A1 (5)
		Total 12 marks

Type MS · Where two lines meet

Answer 4 of 4 · Question 9 part a · 7 marks

2022 JUN · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	**NOTE** ; The algebra must be correct for the first three marks.	
	B1	For writing an equation for L_1 e.g., $y-7=m(x-4)$ **OR** Uses $y=mx+c \Rightarrow y=mx+(7-4m)$
	M1	For the gradient of the perpendicular $\left(-\frac{1}{m}\right)$
	M1	For writing an equation for L_2 (ft a changed gradient) e.g., $y-k=-\frac{1}{m}(x-4)$ **OR** Uses $y=mx+c \Rightarrow y=-\frac{1}{m}x+\left(k+\frac{4}{m}\right) \Rightarrow \left[y=\frac{-x+km+4}{m}\right]$
	M1	For rearranging their equation of L_2 to make x the subject e.g., $x=\frac{y-7}{m}+4$ or $x=km-ym+4$ **ALT 1** Rearranges either equation to make $(x-4)$ the subject $x-4=\frac{y-7}{m}$ and/or $x-4=ym+km$ **ALT 2** Eliminates y from their equations to obtain $'x-4m+7'='\frac{-x+4+km}{m}' \Rightarrow x='\frac{4+km+4m^2-7m}{m^2+1}'$ They must reach $x=\ldots\ldots$ for the award of this mark. **Allow sign errors/slips at this stage.**
	M1	Forming a linear equation in y No simplification is required at this stage. $y-k=-\frac{1}{m}\left(\frac{y-7}{m}+4-4\right)$ **ALT 1** $\frac{y-7}{m}=ym+km$ **ALT 2** Forms a linear equation in y using their expression for x by substituting back into either L_1 or L_2 **Allow sign errors/slips at this stage.**
	dM1	For attempting to rearrange to obtain the required result. **Allow sign errors but there must be no missing terms. You must check carefully.**
	A1	For the required result with no errors seen. $Y=\frac{7+m^2k}{m^2+1}$ allow $y=\frac{7+m^2k}{m^2+1}$

Type MS · Where two lines meet

Answer 4 of 4 · Question 9 part a · 7 marks

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(b)	**B1**	For identifying y coordinate of the midpoint of AB $\left([4], \frac{k+7}{2}\right)$ the x coordinate is not required for this mark.
	M1	Equating the y coordinate of C and y coordinate of the midpoint to give $\frac{k+7}{2} = \frac{7+m^2k}{m^2+1}$
	M1	For rearranging and factorising to attempt achieve $m^2(k-7) = k-7$ This mark can implied by the correct value for either m^2 or m
	A1	Concludes that $m = \pm 1$ $\left[m^2 = \frac{k-7}{k-7} = 1 \Rightarrow m = \pm 1\right]$ Again, this can be implied by the correct $m = -1$
	A1	For $m = -1$
	ALT 1	
	B1	Let coordinates of C be (X, Y) Sets: length AC = length BC $(X-4)^2 + (Y-7)^2 = (X-4)^2 + (Y-k)^2 \Rightarrow \left[(Y-7)^2 = (Y-k)^2\right]$
	M1	Using the coordinates for C of (X, Y) or otherwise, performs the following algebraic manipulation. $(Y-7)^2 = (Y-k)^2 \Rightarrow Y^2 - 14Y + 49 = Y^2 - 2kY + k^2 \Rightarrow -14Y + 49 = -2kY + k^2$ $\Rightarrow 2Y(k-7) = (k+7)(k-7)$ $\Rightarrow 2Y = k+7$ **NB: Allow sign slips, but not missing terms**
	M1	Substitutes in the given expression for Y and attempts to obtain $m^2(k-7) = k-7$ $\frac{7+m^2k}{m^2+1} = \frac{k+7}{2}$ $\Rightarrow 14 + 2m^2k = km^2 + 7m^2 + k + 7 \Rightarrow 7 + km^2 = 7m^2 + k$ $\Rightarrow m^2(k-7) = k-7$ **NB: Allow sign slips, but not missing terms**
	A1	Concludes that $m = \pm 1$ $m^2 = \frac{k-7}{k-7} = 1 \Rightarrow m = \pm 1$
	A1	For $m = -1$
	ALT 2	
	If candidates deduces that $BAC = 45°$ and concludes that the gradient must be $m = -1$ award B1M1M1A1A1 If they leave their answer as $m = \pm 1$ or just $m = 1$, withhold the final A mark	

Type MS · Where two lines meet

Answer 4 of 4 · Question 9 part a · 7 marks

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Question number	Scheme	Marks
	$\log_2 x$ $\log_2 x$	